Unsupervised Learning and Dimensionality Reduction Report

Student
Georgia Institute of Technology
GT ID:

I. Introduction and Datasets

A. Introduction

This report examines the results of a series of experiments on Unsupervised Learning techniques, both standalone and in relation to the downstream task of training Neural Net classifiers. The aim is to provide a comprehensive analysis of how they perform, both quantitatively and qualitatively, and explain their success and failures in relation to existing literature, course material, and quantitative measurements. Both datasets examined are tabular but present very different characteristics outlined in the next sections. Those will inform our hypotheses, which we will revisit throughout the report and in its conclusion, to determine whether they can be accepted or not, based on the data at hand.

B. Hotel Dataset

This dataset comprises 119,390 observations spanning 32 features about booking data from two hotels [1]. Its pre-processing pipeline had been already prepared for the SL and OL reports [2], [3], which we enhanced with the inclusion two new steps tailored for distance-based algorithms (Tab. I). We did not use k-fold cross validation and instead relied on the same time aware train/val/test splits used for the OL Report. Train and test were divided in a 80/20 proportion and further 20% of the training set was reserved as a validation holdout, as described in Table II. Model selection experiments were fitted on the training set and used the holdout set to validate hyper-parameters and model complexity. The final model was fit on the full training set, including the validation holdout. As for the test set, even if this project entailed downstream tasks only for the US Accident, we followed proper scope hygiene practices and reserved it for any future tasks of such nature.

C. Accidents Dataset

This dataset is much more complex and varied; its original version includes 7,728,394 observations across 45 columns [4]. We used the 2M version, as justified in the next section, which was provided by the Teaching Staff as a stratified subsampling of the original data. The split strategy was the same as for the Hotel dataset and again consistent with the previous report, with time aware train/val/test splits used in the same way: train and separate validation holdout for model selection, and train and val combined for fitting the final model and evaluating its performance. The test set was again left untouched and reserved for the downstream task of Neural Net Classifier training. The pre-processing pipeline followed almost identical steps compared to the Hotel one, barring the obvious differences in the very features involved. Hyperparameter selection, described in detail in each algorithm-specific section, ran on a geographically stratified subset of 10% of training data, with the same time-aware split proportions. In contrast, complexity analyses to determine the optimal cluster/feature space size ran on the full dataset, to ensure maximum validity and generalization.

Step	Aim	Change compared to OL
Leakage check	Drop columns that predict or leak the target	Kept for consistency with downstream task, not stricly necessary for UL
Cyclical encoder	Encode cyclical features as month, day of the week, hour in polar coordinates (sin,cos)	New, to allow meaningful structure capturing for UL methods on cyclical fields
Date handling	Convert categorical dates in numeric format	Unchanged
Imputer	Impute missing values	Unchanged
Categorical encoder	Encode categorical data	From Target to Frequency encoders for high cardinality features (label free)
Variance filter	Drop near-zero variance features	New, avoids numerical issues in matrix inversions
Scaler	Scale data	Replaced Standard with Robust Scaler with 1%-99% outlier clipping to avoid outliers dominating distance metrics

TABLE I
Preprocessing steps, common to both datasets, and changes with respect to the OL report.

D. Hypotheses

Based on the domain knowledge about these two datasets accrued during the past SL and OL Reports, and based on the characteristics of the algorithms used throughout this report, I expect (H1) Randomized Projections to struggle capturing meaningful structure for the US Accidents dataset, which features complex decision boundaries and severe temporal effects (see [3], [2]), and to a lesser extent with the Hotel Dataset, due to its sample/feature ratio. In fact, the dimensionality bound formulated in the Johnson-Lindenstrauss Lemma [5] to ensure an ε-bounded distortion would entail projecting more features m than are even available: $m \geq 4 \cdot \ln(N)/(\varepsilon^2/2 - \varepsilon^3/3)$. Even if this bound has been proven to be overly cautious [6], posing for example N = 2,000,000 and $\varepsilon = 0.1$ yields $m \geq 12435.99$, which is many more features than we have. The second hypothesis (H2) is that GMM will outperform kMeans on both datasets, due to its ability to produce elongated clusters more suitable for

the complex decision boundaries featured in our datasets. Finally, with respect to the downstream task of Neural Net classification, based on previous results [3],[2] I expect (H3) speed improvements rather than performance, since our baseline net exhibited underfitting due to its size, depth and limited number of epochs. Finally, due to the complex decision boundaries involved in both datasets, I expect little to no separation of the target variable in the clustering attempted (H4), which we will measure through ARI and MCI. We will revisit these hypotheses throughout the paper and in the Conclusion.

E. Compute Justification and final splits

During this project, we worked with the same setup than in the OL Report: Colab Pro with a 50GB-RAM Tesla GPU instance (15GB VRAM). The CPU is a quad-core Intel(R) Xeon(R) CPU @ 2.20GHz. We worked with efficient dtypes, cached the preprocessed datasets in memory, and worked with sparse representations where possible. We elected to work on the 2M dataset to allow for deeper experimentation and faster model iterations, especially given the computational load posed by the ICA algorithm. All scripts ran in Python, with SkLearn as the library providing the main algorithms. In addition to splits described in

Split	# samples	Pos. prev
Hotel TRAIN	76409	36.21%
Hotel VAL	19103	35.59%
Hotel TEST	23878	40.86%
US Acc. TRAIN	1280000	26.19%
US Acc. VAL	320000	9.92%
US Acc. TEST	400000	5.70%

TABLE II

DETAILED SPLIT SUMMARY PER DATASET.

Table II, we applied some caps to an exploratory study: the trustworthiness metric, measured in experiments sensitivity to noise, was restricted to 20,000 rows, and tSNE and DBSCAN informative 2D plots were generated on a random sample of 30,000 units. Furthermore, we did explore the use of Silhouette scores, which has been shown to scale down remarkably well to small random samples without meaningful precision degradation [7], but in order to avoid possibly biased sub-samplings we chose the Davies–Bouldin index as our structural metric of choice and computed it on the full dataset, together with the Calinski-Harabasz scores. Hence, all model fittings, transformations, and metrics in this report use the dataset in its complete form.

II. DIMENSIONALITY REDUCTION

A. Setup and common settings

This study followed three phases: hyperparameter sweeps at a fixed number of features; complexity study given the optimal knobs determined in the previous phase, and the final dimensionality reduction determined in the previous two stages. The metric of choice during model selection and evaluation was cumulative variance for PCA, mean absolute kurtosis for ICA and reconstruction error via pseudo-inverse for RP. All grids are noted in Table III. The final fitting and data reduction were ran three times with different random seeds {42, 100, 999}, whereas model selection ran on the first seed only {42} to allow for broader exploration within time limits. The condition number of this dataset is 693.945, suggesting the presence of many redundant features. This seems plausible, as there are many overlapping fields regarding dates, features as "babies", "children" which might be overlapping, and others.

Algorithm	Parameter	Grid
PCA	svd_solver	[auto', 'full' (← Hotel only), 'covariance_eigh', 'arpack', 'randomized']
ICA	whiten	['arbitrary-variance', 'unit-variance', False]
	max_iter	Hotel: [200, 300] Accidents: [300, 500]
	tol	[1e-4, 1e-5]
PCA/ICA/RP	n_components	Hotel:[2, 3, 4, 6, 8, 10, 12, 15, 18, 20, 25, 30] Acc:[2, 4, 6, 8, 10, 12, 15, 20, 30, 40, 50, 60]

TABLE III

DIMENSIONALITY REDUCTION - HYPERPARAMETER SEARCH GRIDS (BOTH DATASETS)

B. Hotel Dataset

Algorithm	Parameter	Value
PCA	n_components	20
	svd_solver	covariance_eigh
ICA	n_components	20 (best 5 selected → 95% ∑|kl)
	whiten	unit-variance
	max_iter	200
	tol	0.0001
RCA	n_components	25

TABLE IV

DIMENSIONALITY REDUCTION - FINAL PARAMETERS (HOTEL)

During the model complexity study we set a specific target for each method: 95% cumulative explained variance for PCA (Fig. 1), knee detection on mean kurtosis, plus post-transformation picking the best components to achieve 95% cumulative abs kurtosis for ICA (Fig. 2) and a rather low target of max. 50% of relative reconstruction error for RP . Monitoring the reconstruction loss across increasingly

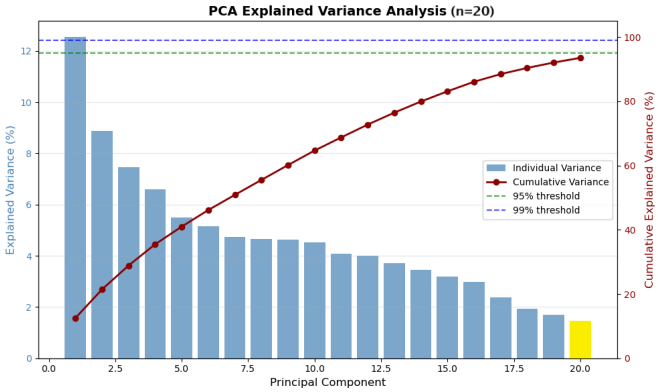

Fig. 1. PCA on Hotel. Cumulative explained variance for n=20

complex random projections resulted in a lower bound of

25 dimensions (Tab IV), which more than the lower bound posed by Larsen and Nelson [6] for our desired $\varepsilon = 0.5$: $\Omega(\frac{log(n_test+val)}{\varepsilon^2}) = \Omega(\frac{log(95512)}{0.5^2}) \approx 19.92$. Still, the bound is asymptotic in N, so this result still adheres to the literature and the Lemma. Inspecting component loadings reveals

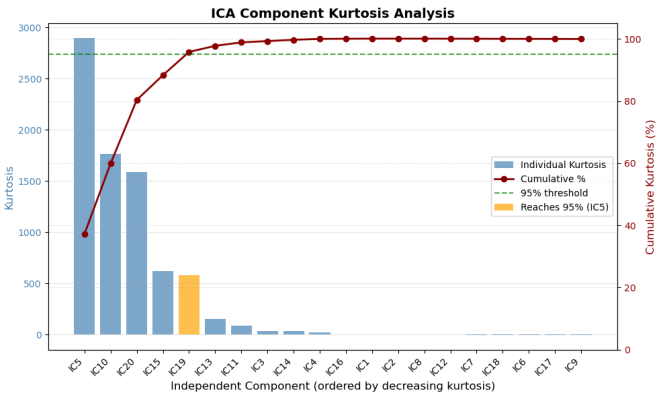

Fig. 2. ICA on Hotel. Individual component kurtosis for n=20 and 95% cumulative |k| threshold.

important insights: ICA struggled to identify meaningfully mixed features. The most kurtotic component (|k|=2894.39) was a combination of -1.17*adr-0.3 cos(arrival month)-0.29 cos(arrival month)+0.3*children, perhaps representing families with seasonal patterns and cheap bookings. Beyond this, all other extracted features had almost unitary weight on one single feature and no other contribution. This is not inherently bad, as it might mean that the whitening step removed all correlation and the remaining features where already independent enough; the projected space seems to have meaningful latent structure, as noted in the exploratory cluster target analysis conducted with DBSCAN in Fig. 3, which naturally formed clusters that identified low (C6=8%) and high cancellation (C1/C5=100%) rates. PCA captured

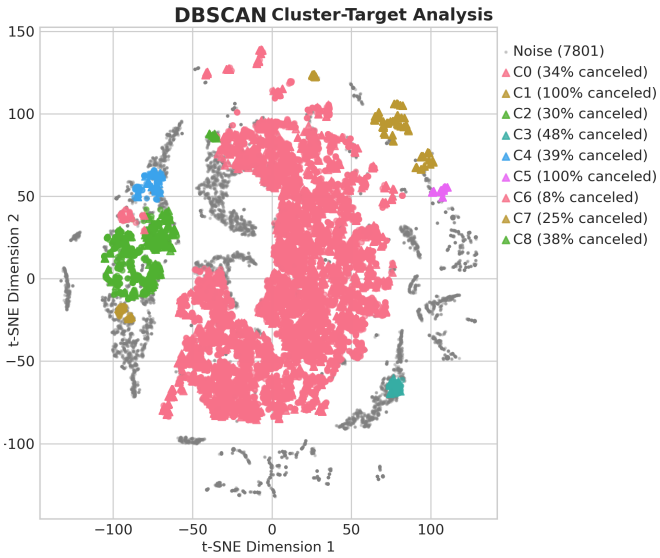

Fig. 3. Explorative DBSCAN clustering on the ICA projected space (2D t-SNE projection for visualization purposes).

instead more nuanced decision boundaries, with the most variance (12.5%) by a single feature again due to seasonality (year date and sin(month)), and the second best focused on expensive and long stays (adr, stays in week nights, stays in weekend nights, special requests). This difference is also seen comparing Fig 2 and 1 where 95% kurtosis is compressed into 5 features, compared to 95% variance in 20 features for PCA. The effective dimensionality of the PCA-reduced dataset [8] was 15.45 ± 2e-15, demonstrating meaningful dimensionality reduction. Randomized Projections found completely different correlations, for example recurrent customers through online channels, and other market segment-related features, yet the reconstruction error was too high to ensure that the original signal was preserved in the projected space, which confirms our initial hypothesis for this dataset (see Tab V). This latter algorithm proves the most stable against zero-centered Gaussian noise of varying intensities (0-1 in 0.1 increments). The ARI metric in Table V is the average of how stable k-Means cluster remain in the projected space at varying levels of noise, compared with the same clusters on the baseline.

Metric	PCA	ICA	RP
Fit time(s)	0.0084 ± 0.0004	0.9046 ± 0.9181	**0.002** ± 0.00004
Reconstruction MSE	**0.0282** ± 0.0	**0.0282** ± 0.0	0.2646 ± 0.0065
Collinearity reduction Ratio	236.0405 ± 3.4809	**693.9357** ± 0.0057	23.8721 ± 6.1998
Clustering ARI stability	0.8322 ± 1.3597e-16	0.1022 ± 5.1290e-06	**0.8506** ± 0.1395

TABLE V

DIMENSIONALITY REDUCTION - METRICS (HOTEL)

C. Accidents Dataset

Algorithm	Parameter	Value
PCA	n_components	40
	svd_solver	auto
ICA	whiten	false (89d kept to achieve 95% |k|)
	max_iter	1000
	tol	0.0001
RCA	n_components	50

TABLE VI

DIMENSIONALITY REDUCTION - FINAL PARAMETERS (US ACCIDENTS)

Exploratory studies yielded higher component numbers for this larger dataset (Tab.VI) and fascinating results. The condition number of the original dataset post-processing is 1125.15, suggesting many redundant or correlated features. Pre-processing pipeline yielded 104 features, since OHE was used sparingly. PCA yielded the required 95% explained variance at 40 components, and component loadings revealed weather-aligned components (See Fig.4). The effective dimensionality of the PCA-reduced dataset [8] was 13.04 ± 2e-15, which validates the efficacy of this method. Whitening policies were explored during the model selection phase, but yielded anomalously large kurtosis values, possibly due to outliers or numerical issues, that should be inspected further. Applying no whitening to such a highly conditioned dataset

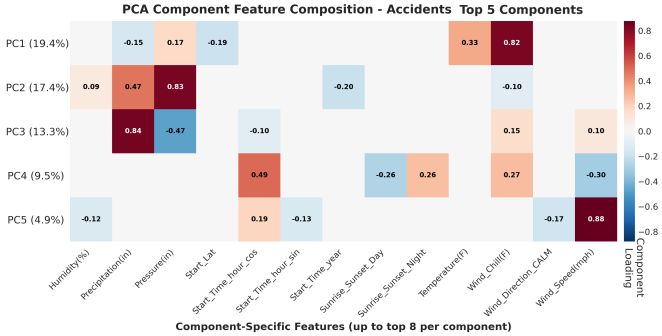

Fig. 4. Component loading heatmap - PCA on US Accidents (seed 42). Top 5 eigenvalues by explained variance and top 8 contributing features shown.

led to reaching $\sum|k| = 95\%$ at 85, 88 and 89 components for seeds 42, 100, and 999 respectively. As such, collinearity reduction was nonexistent and reconstruction error too (Tab. VII), since almost all dimensionality was kept. The most kurtotic features had a value of |k| in the range of 3-5, and were mostly location-based, including "State" OHE features and some weather conditions. For RCA, the 50% ε target was met at 50 components. Component loadings revealed weather features again, together with road features (e.g. crossing, junction) and State-based OHE (e.g. State_CA). The projection looks promising and meaningful domain-wise, but the high reconstruction error suggests that it will pose problems in downstream SL use VII and points towards our initial hypothesis (H1, Section I-D).

Metric	PCA	ICA	RP
Fit time(s)	0.2912	1177.3522	**0.0609**
	± 0.0446	± 7.4278	± 0.0022
Reconstruction	0.0378	7.7e-13≈**0**	0.5579
MSE	± 0.0	± 6.5e-14≈0	± 0.0157
Collinearity	**116.1444**	1.55e-05	26.4506
reduction Ratio	± 0.0	± 1.96e-07	± 3.1299
Clustering	**0.3083**	0.1720	0.0197
ARI stability	± 1.7e-08	± 0.01197	± 0.02633

TABLE VII

DIMENSIONALITY REDUCTION - METRICS (US ACCIDENTS)

III. CLUSTERING

A. Setup and common settings

An initial hyperparameter sweep focused on hyperparameters at a fixed number of clusters, followed by a focused complexity study with the best parameters set, and finally the best version of the model was fit on the data. Table VIII details the parameters tested. The metric of choice is the Davies-Bouldin index, which accounts for how intra-cluster similarity and infra-cluster distance, defined as follows, where S_i is the avg distance between the center of a cluster and its elements (Eq. 1,2). This metric is not normative, which means there is not an absolute range of "good" or "bad" values [9], yet it is fast and reliable for comparing clustering results across the same domain.

$$DB(k) = \frac{1}{k}\sum_{i=1}^{k} \max_{i \neq j} \left\{ \frac{S_i + S_j}{d(\bar{x}_i, \bar{x}_j)} \right\} \tag{1}$$

$$S_i = \frac{1}{n_i}\sum_{j=1}^{n_i} d(x_j, \bar{x}_i) \tag{2}$$

Algorithm	Parameter	Grid
K-means	init	[k-means++,random]
	tol	[1e-3,1e-4]
	max_iter	[300]
	n_init	Hotel:3, Accidents:10
GMM	covariance_type	[tied,diag,spherical]
		Round II: best vs full (only for Hotel)
	init_params	Hotel: [random_from_data,kmeans]
		Acc.: [k-means++,random_from_data]
	tol	[3e-4]
	reg_covar	[1e-3, 3e-4, 1e-4]
	n_init	Hotel:3, Accidents:10
K-means/GMM	n_components	Hotel:[2,3,4,6,8,10,12,15,18,20,25,30]
		Acc: [2,4,6,8,10,12,15,20,30,40,50,60]

TABLE VIII

HYPERPARAMETER GRIDS - CLUSTERING

B. Hotel Dataset

Complexity analyses were carried out with the knee detection heuristic, using DB score, which led to the clustering described in Tab. IV. Hyperparameter exploration indicated

Algorithm	Parameter	Value
K-Means	n_clusters	6
	init	random
	tol	0.001
	max_iter	300
	n_init	5
GMM	n_components	8
	covariance_type	spherical
	init_params	kmeans++
	tol	0.0003
	reg_covar	0.0001
	max_iter	300
	n_init	5

TABLE IX

CLUSTERING - FINAL PARAMETERS (HOTEL)

that spherical covariance provided the best DB scores on the validation set; we tested the more costly "full" covariance against "spherical", but found the latter to be superior, given the data and task at hand. Metric wise, k-Means proved an order of magnitude faster than GMM, and better at providing structural separation among clusters (Tab. X), with both method falling short of separating the targets meaningfully (see almost zero ARI and Normalized Mutual Information in Tab. X), confirming hypothesis H4 (Section I-D). A visual exploration surprisingly identifies k-Means as more able to capture elongated clouds of positive samples (see #3 in Fig. 5, 6), even if GMM shows more cluster separation due to a higher k. Still, k-Means proves quantitatively better at a fraction of the time and compute cost. This contradicts my initial hypothesis at least for this dataset (H2, Section I-D).

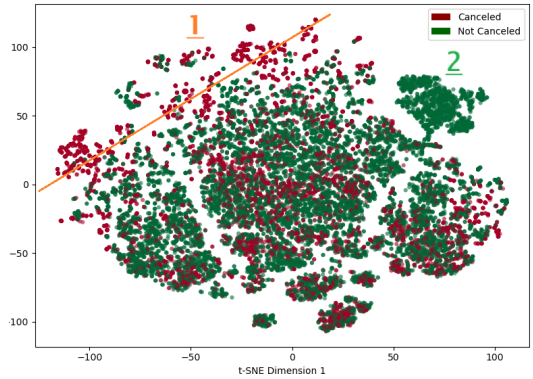

Fig. 5. tSNE 2D projection of the Hotel dataset, with 2 markers for comparative analysis

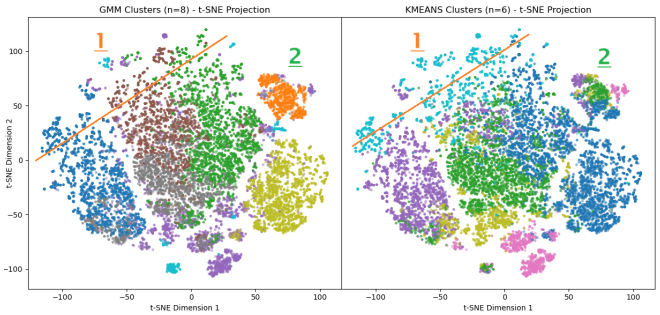

Fig. 6. Comparative analysis of kMeans and GMM on Hotel dataset - projected in 2D via tSNE (for illustrative purposes only)

Metric	k-Means	GMM
Fit time(s)	**0.4846**	6.9842
Iterations	42	**24**
Davies B.	**2.4702**	2.9035
Calinkski-Har	**5477**	4310
Inertia	1588315	-
BIC	-	7136808
AIC	-	7132955
ARI	**0.0305**	0.0023
NMI	**0.0341**	0.02715

TABLE X

CLUSTERING - METRICS (HOTEL)

C. Accidents Dataset

For this more complex dataset, complexity analyses showed a knee with diminishing returns in term of DB score at 6 clusters for both algorithms (Tab. XI), and the performance gap was smaller, yet k-Means again proved superior and much faster (Tab. XII). Indeed GMM with spherical covariance could not express its full geometrical flexibility, but other covariance types did not reach the same DB score during model selection and were hence discarded. The full covariance was not attempted, due to its high computational cost. In conclusion, quantitative results contradicted hypothesis H2 formulated in the introduction for this dataset too. GMM exhibited relatively higher, but still close to zero values of ARI and NMI, in line with out hypothesis H4 (Section I-D) of clustering algorithms failing to naturally align with areas of same target features.

Algorithm	Parameter	Value
K-Means	n_clusters	6
	init	k-means++
	tol	0.001
	max_iter	300
	n_init	5
GMM	n_components	6
	covariance_type	spherical
	init_params	kmeans
	tol	0.0003
	reg_covar	0.0001
	max_iter	300
	n_init	5

TABLE XI

CLUSTERING - FINAL PARAMETERS (US ACCIDENTS)

Metric	k-Means	GMM
Fit time(s)	**16.1869**	104.3695
Iterations	**10**	13
Davies B.	**1.7014**	1.7133
Calinkski-Har	**258299**	255509
Inertia	16496150	-
BIC	-	88717439
AIC	-	88709640
ARI	-0.0013	**0.0026**
NMI	0.0082	**0.0093**

TABLE XII

CLUSTERING - METRICS (US ACCIDENTS)

IV. CLUSTERING ON REDUCED DATA

A. Setup and common settings

For this experiment, we chose to kept the same best hyper-parameters determined in Tab. IX, XI. The reason for this is being able to assess and isolate more clearly those differences due to a simplification of the underlying manifold/feature space, without introducing variations in the inner workings of the clustering itself. This led to experiments (see Tab.XIII) where we projected up from n features to m>n clusters, which we will examine in detail. As a general rule, during the initial clustering process, if across the three seeds we had obtained different dimensionalities (as happened for ICA), we clustered the highest dimensionality version, with the rationale of retaining as much information as possible.

B. Hotel Dataset

Combination (dims)	Fit time(s)	Davies B.	Calinski	NMI	BIC
PCA(20)→Kmeans(6)	**0.3093**	**2.4312**	**5989**	0.03189	-
PCA(20)→GMM(8)	8.9547	2.7788	4744	**0.03193**	4576140
ICA(5)→Kmeans(6)	**0.2175**	**0.9244**	**19167**	**0.0051**	-
ICA(5)→GMM(8)	6.7903	3.0409	6926	0.0092	261684
RP(25)→Kmeans(6)	**0.4495**	**2.0192**	**8055**	**0.0326**	-
RP(25)→GMM(8)	4.8597	2.6556	5871	0.0272	5346173

TABLE XIII

CLUSTERING ON DIMENSIONALITY REDUCED SPACES - METRICS (HOTEL). DIMENSIONALITY AND CLUSTERS NOTED IN PARENTHESES.

On all projected spaces, K-means systematically outperformed GMM, in particular on the low-dimensional ICA space (5 features), probably due to GMM over-clustering (8 clusters vs. 6 for K-means). Fit time was consistently 10-20

times faster for K-means, and both methods failed to capture almost any shared information about the target within their clusters (NMI in Tab.XIV), as hypothesized (H4) in Section I-D. The most well-separated clustering was produced by K-means on the ICA projection, with DB=0.9244. Furthermore, the low-dimensional ICA projection seems to have distorted the original signal encoded in the data, since no pair of different algorithms/reduced spaces agreed almost on no cluster assignments (Fig. 7) even if the ICA space was the smallest (5d).

C. Accidents Dataset

Combination	Fit time(s)	DB sc.	Cal sc.	NMI	BIC
PCA(40d)					
→Kmeans(6)	**8.3041**	**1.6227**	**281403**	0.0083	-
→GMM(6)	72.6596	1.6299	278230	**0.0096**	92,5e6
ICA(89d)					
→Kmeans(6)	**14.2072**	**1.4869**	**415534**	**0.0151**	-
→GMM(6)	82.4533	1.5302	407370	0.0150	-14,86e6
RP(50d)					
→Kmeans(6)	**9.9219**	**1.6388**	**288997**	0.0097	-
→GMM(6)	52.740	1.6651	285446	**0.0099**	101,91e6

TABLE XIV

CLUSTERING ON DIM. REDUCED SPACES - METRICS (US ACCIDENTS). DIMENSIONALITY AND CLUSTERS NOTED IN PARENTHESES.

US Accidents results show much more uniform results, both in terms of structural performance (see DB and Calinski scores in TabXIV) and cluster agreement (Fig 7, right). ICA appears notably different than PCA and RCA (slower and in disagreement) due to very conservative feature reduction (104→89=↓14%) in comparison to PCA and ICA (see Fig. 7 and Tab XIV) and is much more similar to the original clustering metrics (Tab. XII). PCA+k-Means proved to be

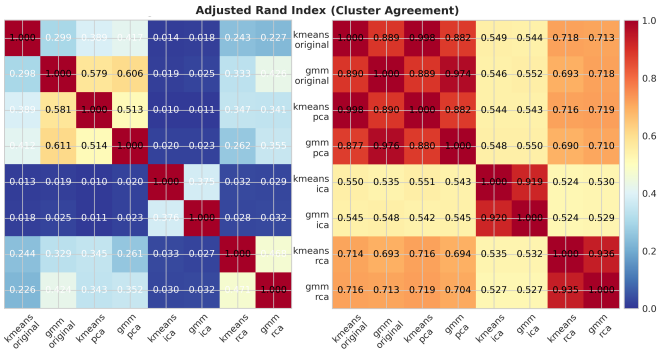

Fig. 7. Adjusted Rand Index (Cluster agreement) across different methods, for both Hotel Dataset (left) and US Accidents (Right)

the best combination, with PCA(40d) providing significant feature compression with low reconstruction error (0.0378, Tab. VII), and k-Means obtaining better DB score compared to k-Means on the full dataset (1.6227 vs 1.7014) in almost half the time (8.3s vs 16.1s), as noted in Tab. XII, XIV. The gap with GMM remained small as in the full version, and RP provided also well separated clusters (DB≈1.6 for both GMM and k-Means), but its high reconstruction error could pose problem in downstream SL tasks, as formulated in H1 (Section I-D).

V. DOWNSTREAM TASKS: TRAINING A NEURAL NET CLASSIFIER

A. Setup, common settings and metric tables

For this section, we identified the best performing Neural Net configuration from the previous report [3], which is a Neural Network with Nesterov Optimizer, no dropout or regularization applied, as we had previously determined that those were detrimental to its performance on the US Dataset. It features five fully connected hidden layers: the first four of size 300x300 and the last sized 200x200. The original task was to classify severe accidents (Severity at onset > 2), and we had previously noted that the dataset exhibited severe temporal effects, with positive ratio noted in Table II declining from 26% (train) to 5% (test). The Nesterov Optimizer operates with a lr=0.3, momentum = 0.7, and batch size of 2048 samples. The learning rate scheduler is CosineAnnealingLR with minimum rate 1e-5 and T_max = 15 epochs, our limit that we kept untouched for consistency with the OL report. The learning criterion is a BCEWithLogitsLoss with weight balancing computed once on the train split.

We list all metrics gathered across both experiments here below, which we will refer throughout this paragraph, in one place for convenience and ease of comparison cross-methods and metrics.

Method	Best AP Epoch	Grad. Evaluations	Fit Time (s)
Baseline	10.7±0.9	9375.0±0.0	245.53±2.54
PCA(40d)	**12.0±0.8**	9375.0±0.0	224.66±3.02
ICA(89d)	13.3±0.9	9375.0±0.0	244.27±1.51
RCA(40d)	14.3±0.9	9375.0±0.0	**224.35±0.93**
KMeans(6)	**9.3±0.5**	9375.0±0.0	**225.96±0.89**
GMM(6)	10.7±1.7	9375.0±0.0	230.76±4.56

TABLE XV

NEURAL NET METRICS - MIXED STRATEGIES (SPEED)

Method	Final Val Loss	Final Tr. Loss	Final Val AP	Best Val AP
Baseline	0.3864±0.0057	0.6490±0.0007	0.5933±0.0028	0.5970±0.0020
PCA(40d)	0.3834±0.0024	**0.6544±0.0010**	0.5899±0.0053	0.5916±0.0041
ICA(89d)	0.4491±0.0063	0.6900±0.0011	0.5286±0.0025	0.5311±0.0056
RCA(40d)	**0.3801±0.0003**	0.6602±0.0013	**0.5981±0.0009**	**0.5983±0.0011**
KMeans(6)	0.3849±0.0037	0.6490±0.0021	0.5935±0.0051	0.5964±0.0022
GMM(6)	**0.3826±0.0062**	**0.6483±0.0018**	**0.5995±0.0036**	**0.6014±0.0032**

TABLE XVI

NEURAL NET METRICS - MIXED STRATEGIES (TRAINING DATA)

Method	Test Loss	Test AUC	Test AP	Test F1
Baseline	0.3126±0.0008	0.8623±0.0008	0.4612±0.0001	0.4901±0.0042
PCA(40d)	**0.3144±0.0015**	**0.8560±0.0012**	**0.4477±0.0010**	**0.4846±0.0024**
ICA(89d)	0.3501±0.0014	0.8317±0.0019	0.3831±0.0139	0.4226±0.0083
RCA(40d)	0.3160±0.0006	0.8549±0.0029	0.4453±0.0050	0.4793±0.0017
KMeans(6)	0.3142±0.0036	0.8593±0.0031	0.4591±0.0028	**0.4927±0.0030**
GMM(6)	**0.3133±0.0026**	**0.8605±0.0016**	**0.4602±0.0006**	0.4922±0.0020

TABLE XVII

NEURAL NET METRICS - MIXED STRATEGIES (TEST METRICS)

B. Reduced data

In this first SL experiment, we trained our baseline Nesterov MLP on the original dataset and on each of the dimensionality reduced versions of the US Accidents dataset. Since we had added two steps to the pre-processing pipeline

(Tab. I), we decided that the baseline model would be trained with the new same pipeline that determined the clusters/reductions. We want to keep our comparative study focused on the methods as much as possible, and free the comparison from any confounding factors, such as different pre-processing steps. Training curves immediately show that

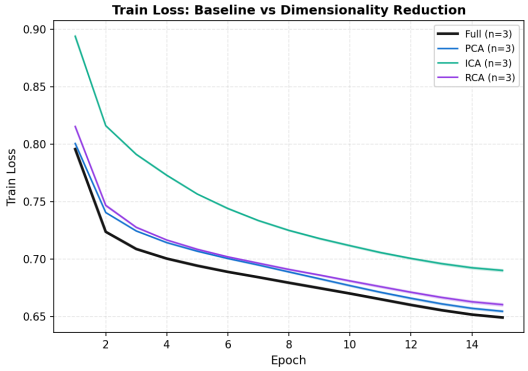

Fig. 8. Train loss curves: baseline vs dimensionality reduction

the model trained on the ICA projection not only did not achieve meaningful compression VII at 89 dimensions, but actually worsened model performance with a much higher train, val and test loss than all other methods (Tab. XV, XVI, XVII, Fig. 8). It suffers in almost all metrics except for

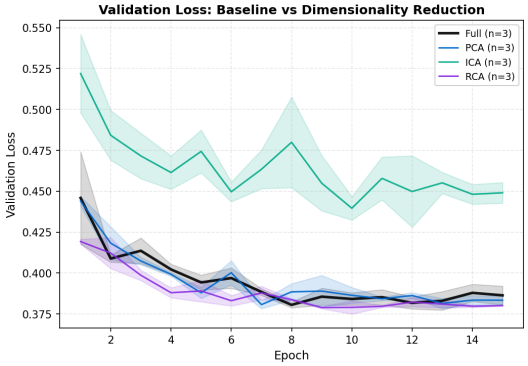

Fig. 9. Validation loss curves: baseline vs dimensionality reduction

AUC, which is naturally high for all models, given the great class imbalance on the test set (95% negatives). Both models trained on PCA and RCA projections achieve better final validation losses compared to the baseline (Tab. XVI, Fig.9) in a similar time of 224s. The model trained on the PCA space though exhibits consistently better test metrics with respect to RCA, but loses to the baseline model. The main metric of AP features a 2.92% decrease between Baseline and PCA but an 8.57% time saving, which seems a moderately good tradeoff. In conclusion, PCA is the clear winner in this cross-dimensionality reduction study, beating RP by a small margin and ICA by a large margin, the latter struggling to achieve comparable results. The very low reconstruction error of PCA VII combined with its collinearity reduction ratio (116.1444) helped the model achieve comparable results

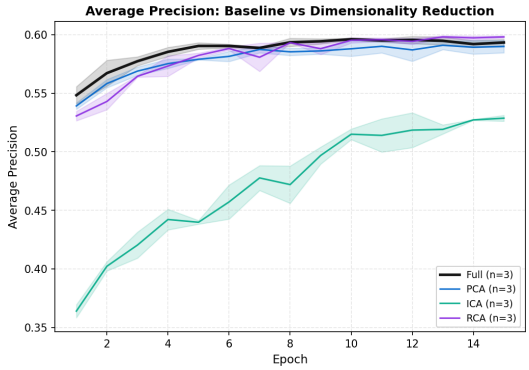

Fig. 10. Average Precision curves: baseline vs dimensionality reduction

in a 8.57% less time, albeit with a ≈3% worse AP on the test set.

C. Clustered data

Given that our model selection study yielded a low number of clusters/components (6) for both methods, we decided that we would include those features mixing them with the original features of the full space. In particular, we added cluster assignments for k-Means, and instead for GMM added soft probabilities, which meant six additional features in both cases. In the former case, we added six binary integers (OHE on cluster assignments), in the latter case we added six continuous features corresponding to the responsibility of each component for that specific data point. Performance on the training set were almost identical for all

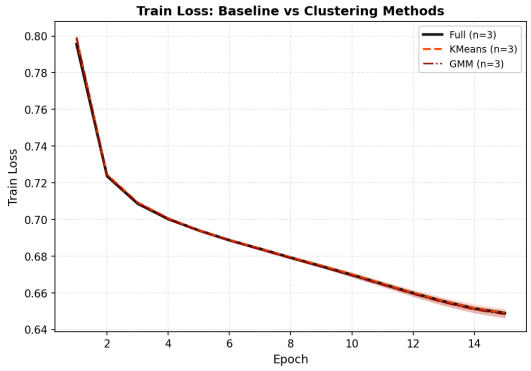

Fig. 11. Train loss curves: baseline vs cluster-enhanced data

three models (baseline, +k-Means, +GMM), both in train loss and train AP (Tab.XVI. On the validation set, both enhanced models outperformed the baseline in less time (15-20 seconds less). A reduction in time complexity was unexpected, since we added more features and performed the same number of budgeted gradient evaluations. It could be due to better-conditioned gradients due to the smoother feature space afforded by the global information encoded in the cluster information (Tab XII suggests well-structured clusters for both methods). The model enhanced with GMM responsibilities beats both the baseline and the k-Means version in most metrics (loses to k-Means in speed and is

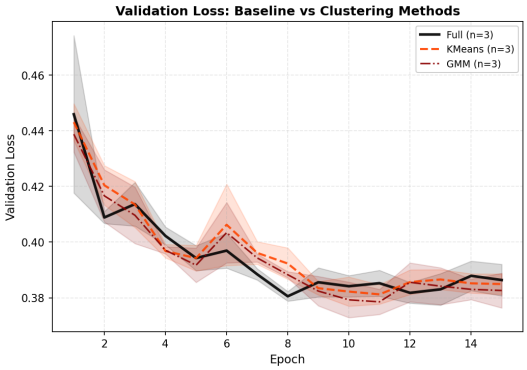

Fig. 12. Validation loss curves: baseline vs cluster-enhanced data

comparable in Test F1, see Tab XV). In particular, the GMM enhancement affords a 6.1% speedup with better validation metrics and comparable or better test metrics.

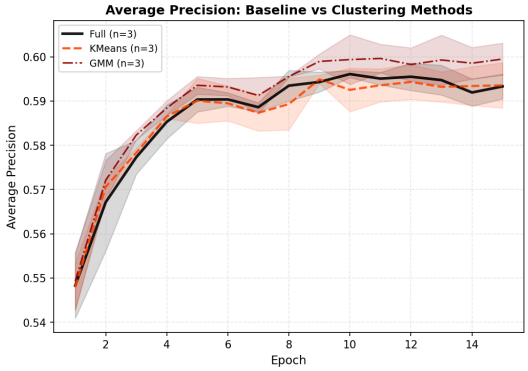

Fig. 13. Average Precision curves (val set): baseline vs cluster-enhanced data

VI. Conclusions

A. Revisiting hypotheses

Throughout this report, we gathered enough data to confirm that indeed Randomized Projections struggled to encode enough information due to the large size of the dataset (H1), in accordance with the Johnson-Lindenstrauss Lemma, as measured in Tab.V (Hotel) and VI (US Accidents). They provided speedups to the downstream task of Neural Net training (Tab.XV) but eventually lost to PCA metric-wise. The initial hypothesis that GMM would have outperformed k-Means (H2) is partially rejected: structurally GMM struggled to provide well-separate clusters (Tab.X, XII), but actually outperformed k-Means in the SL downstream task (Tab XVII). This is rather due to the naturally higher entropy encoded in its continuous responsibilities, and does not seem directly related to a larger structural soundness, also considering that GMM lost to k-Means on all dimensionality reduced datasets too (DB Scores Tab. XIII, XIV). Our instrumentation confirmed that (H4) all clustering failed to naturally group or gather target information (ARI/NMI in Tab.X, XII, XIII, XIV). Finally, we reject the initial hypothesis (H3) that the only improvement afforded by mixing reduction and clustering with Neural Net training would have been speed, as instead both PCA and GMM meaningfully improved both speed and real life-relevant metrics (AP, Tab. XVI). This led to lower Type I errors (higher AP).

B. Decision rules

To summarize our quantitative findings, prefer PCA as a reduction technique on Hotel, due to its low reconstruction error, high tolerance to noise (Tab V) and k-Means as a clustering technique, since it produces well-separated clusters in time one order of magnitude lower (Tab X). Search within a low number of cluster (See lower end of grid in Tab VIII, ideally 2-10) to avoid cluster overcrowding (see GMM k=8 lower metrics, Tab X. For US Accidents, prefer PCA too, as it beats ICA and RP both structurally (ARI stability and collinearity reduction, Tab VII) and in downstream SL tasks (Tab. XVII) achieving comparable if not better metrics at a lower time cost (Tab. XVI).

C. Next Steps

Given the limited success with dimensionality reduction but the encouraging results of its combination with downstream tasks, and the inherent complexity of the US Accidents dataset, the next step is experimenting with Manifold Learning algorithms, such as UMAP or ISOMAP. Those could provide a higher level of flexibility and unveil latent structure in the feature space, providing higher returns in downstream tasks.

AI Use Statement

For this report, I used Claude Sonnet 4.5 through Copilot as a support for debugging and writing blocks of code, as detailed, attributed, and annotated in the codebase (see Git), and ChatGPT5 as a search engine to find peer-reviewed sources on the Johnson-Lindenstrauss Lemma and DB score.

References

[1] N. Antonio, A. de Almeida, and L. Nunes, "Hotel booking demand datasets," *Data in Brief*, vol. 22, pp. 41–49, 2019. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2352340918315191

[2] P. Ferragamo, "Supervised learning report." [Online]. Available: https://www.overleaf.com/read/vnykhqhqkvdh#f9152e

[3] ——, "Optimization learning report." [Online]. Available: https://www.overleaf.com/read/xcmzxkbvysss#d2ef19

[4] S. Moosavi, M. H. Samavatian, S. Parthasarathy, R. Teodorescu, and R. Ramnath, "Data: Us accidents (since 2016) cc by-nc-sa 4.0," 2023. [Online]. Available: https://www.kaggle.com/ds/199387

[5] S. Dasgupta and A. Gupta, "An elementary proof of a theorem of johnson and lindenstrauss," *Random Struct. Algorithms*, vol. 22, no. 1, p. 60–65, Jan. 2003. [Online]. Available: https://doi.org/10.1002/rsa.10073

[6] K. G. Larsen and J. Nelson, "Optimality of the johnson-lindenstrauss lemma," in *2017 IEEE 58th Annual Symposium on Foundations of Computer Science (FOCS)*, 2017, pp. 633–638.

[7] J. Pavlopoulos, G. Vardakas, and A. Likas, "Revisiting silhouette aggregation," in *Discovery Science: 27th International Conference, DS 2024, Pisa, Italy, October 14–16, 2024, Proceedings, Part I*. Berlin, Heidelberg: Springer-Verlag, 2025, p. 354–368. [Online]. Available: https://doi.org/10.1007/978-3-031-78977-9_23

[8] L. L. Campbell, "Exponential entropy as a measure of extent of a distribution," *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, vol. 5, no. 3, pp. 217–225, 1966.

[9] F. Ros, R. Riad, and S. Guillaume, "Pdbi: A partitioning davies-bouldin index for clustering evaluation," *Neurocomputing*, vol. 528, pp. 178–199, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0925231223000528